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Research paper

Interpretable machine-learning framework for engineering optimization of microalgal biofuel yield across cultivation conditions

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ABSTRACT

Optimizing microalgal biofuel yield remains a highly complex engineering challenge due to the intricate, non-linear dependencies among diverse cultivation environments and biological parameters. To provide actionable decision support for harvest scheduling and bioreactor control, this study leverages a comprehensive dataset of 2950 fully retained experimental records, utilizing statistical leverage diagnostics strictly for homogeneity verification without excluding any operational extremes. By evaluating multiple predictive architectures through rigorous independent testing, the Random Forest model demonstrated superior generalization capability, achieving a test-set R^2 of 0.918 and an average absolute relative error (AARE) of 29.7%. Advanced SHAP further revealed that time and type fundamentally drive predictive variance, while highlighting critical non-linear engineering trade-offs regarding optimal nutrient dosing and light intensity. To ensure robust generalization and mitigate data leakage, a rigorous Leave-One-Source-Out (LOSO) cross-validation strategy was implemented. This grouped validation confirmed that the Random Forest model maintained the highest predictive robustness across independent external data sources.

1. Introduction

The continuous escalation of global energy demands combined with the rapid exhaustion of fossil fuel reserves necessitates a critical transition toward renewable energy alternatives [1]. Among the various renewable pathways, third-generation biofuels derived from microalgae have emerged as highly promising candidates. Unlike first- and second-generation biofuels, microalgae do not compete with food crops for arable land, exhibit exceptionally high photosynthetic efficiency, and can accumulate significant intracellular lipids [2]. Furthermore, these microorganisms can be cultivated in marginal environments, including wastewater, highlighting their immense potential for sustainable and environmentally friendly bioenergy generation [3].

Despite the theoretical advantages, the commercialization of algal biofuels is persistently hindered by the complex, non-linear dynamics of biological cultivation. The growth rate and lipid accumulation of microalgal species are heavily dictated by a highly interacting matrix of environmental factors, most notably light intensity, cultivation temperature, medium pH, and macro-nutrient concentrations [4]. Because continuously controlling these operational parameters in photobioreactors or raceway ponds demands substantial energy and capital, identifying the exact operating windows that optimize biological output without incurring prohibitive costs remains a primary engineering challenge [5].

To address the high-dimensional challenges of biorefinery optimization, AI and ML implemented on renewable energy sector [6]. Traditional empirical methods for optimizing biological processes are time-consuming and often fail to capture the complex biological

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synergies occurring within cultivation systems. In contrast, predictive algorithms can rapidly process historical data, allowing for in silico testing and the swift identification of optimal cultivation pathways, thereby bypassing laborious trial-and-error laboratory experiments [6].

Recent literature demonstrates the growing success of machine learning in specifically predicting microalgal yields and optimizing cultivation strategies. For instance, regression models have been successfully utilized to predict final biomass yields early in the cultivation cycle based on optical density and pH, allowing operators to implement corrective measures mid-process [4]. Similarly, advanced ensemble models, such as random forests trained on extensive multi-variable databases, have been deployed to unlock high lipid productivity by identifying critical, previously underappreciated triggers like exogenous iron and phosphorus [7].

Beyond upstream cultivation, computational modeling has also revolutionized the evaluation of downstream processing and fuel quality. Algorithms such as artificial neural networks have been rigorously compared against linear regressions to predict the energetic heating value of algal biomass [8]. Additionally, machine learning has proven invaluable in optimizing the thermochemical conversion of biomass, such as employing gradient-boosted decision trees to maximize fatty acid methyl ester yields during catalytic transesterification [9] and utilizing Gaussian processes to predict moisture kinetics during energy-intensive vacuum drying [10].

Despite these significant advancements, critical shortcomings persist in the existing literature, severely limiting the industrial deployment of predictive models. Most previous modeling efforts strictly apply algorithms to single algal species or highly isolated, homogenous cultivation conditions [7]. Consequently, these models lack the generalized robustness required for application across diverse, heterogeneous biorefinery environments. Furthermore, models trained on literature-derived data frequently rely on standard random cross-validation, a method highly susceptible to data leakage when data points from the same experimental source share hidden confounding variables.

Crucially, the current literature often prioritizes pure statistical accuracy over practical engineering value. An algorithm that perfectly predicts biomass but fails to provide actionable operating ranges, uncertainty intervals, or resource-tradeoff analyses offers limited utility to a bioreactor operator. There is a distinct absence of comprehensive, deployable decision-support frameworks that dynamically link real-time operational inputs to interpretative, species-level process-control rules. To bridge the gap between theoretical data science and practical biorefinery operations, predictive modeling must directly address the cost-benefit realities of continuous cultivation.

Developing reliable, data-driven strategies for microalgal cultivation is absolutely essential to overcoming the economic barriers currently preventing large-scale biofuel commercialization. The engineering problem addressed in this work is the rapid prediction of daily biofuel yield from measurable cultivation variables to support harvest scheduling, environmental control, and resource allocation in microalgal bioreactors. Therefore, this study asks: Which cultivation conditions should be selected to maximize daily yield while limiting energy and nutrient costs? The core hypothesis is that an ensemble-based machine learning framework, validated through strict source-grouped methodologies to prevent data leakage, can accurately model these non-linear biological relationships better than traditional response-surface methods. As illustrated in Fig. 1, the research workflow systematically transitions from the aggregation of a novel, multi-species benchmark dataset to data preprocessing, followed by rigorous algorithm benchmarking, and finally, the extraction of actionable process-control rules. The ultimate objective is to deliver a highly interpretable, field-deployable predictive framework that directly translates continuous sensor data into dynamic optimization strategies for commercial biofuel production.

2. Overview of machine learning algorithms

This study employs a robust and diverse suite of seven advanced machine learning algorithms to construct the predictive framework for the targeted engineering system. The selected models encompass a wide range of learning paradigms, specifically including Decision Tree (DT), Adaptive Boosting (AB), Random Forest (RF), Ensemble Learning (EL), Convolutional Neural Network (CNN), Support Vector Regression (SVR), and Multilayer Perceptron Artificial Neural Network (MLP-ANN). By deploying this comprehensive array of techniques, the study ensures a rigorous comparative analysis capable of capturing the highly non-linear and complex relationships inherent in the bioprocess dataset.

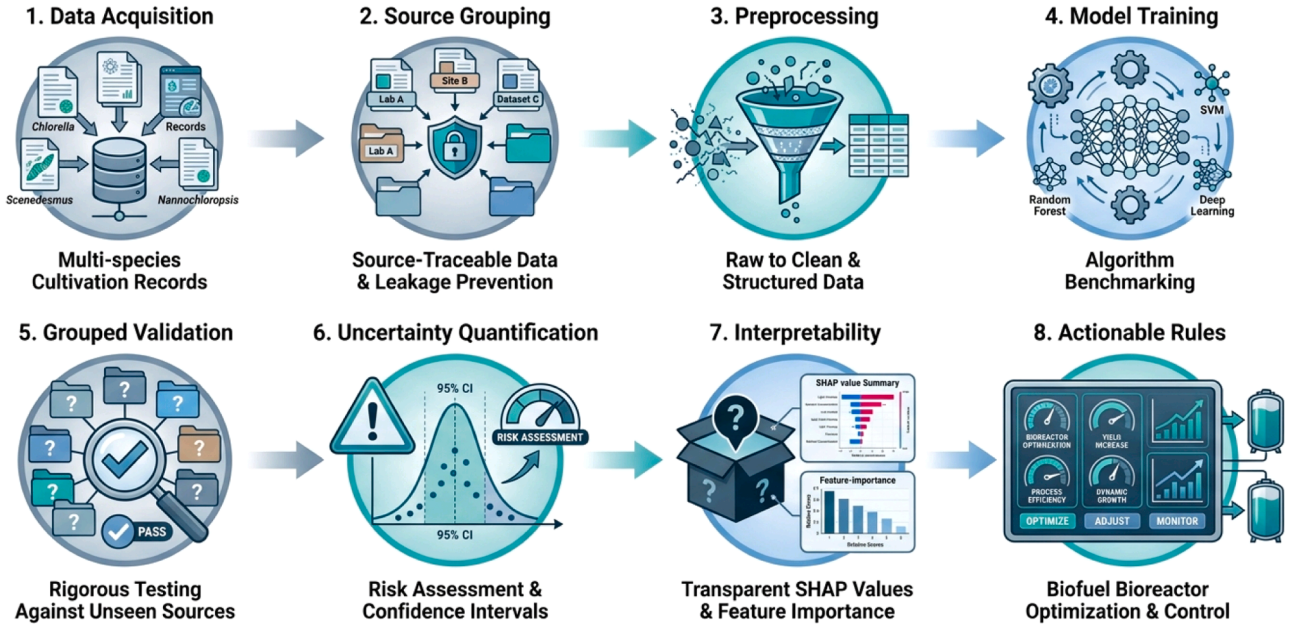


Fig. 1. Schematic workflow of the proposed predictive engineering framework, detailing the progression from source-traceable data acquisition and grouped validation to the extraction of actionable process-control recommendations.

To ensure the highest standards of reproducibility and robust model validation, the implementation of these algorithms followed a strictly standardized pipeline. Following rigorous data preprocessing the performance of each machine learning model was systematically evaluated utilizing a 5-fold cross-validation strategy to mitigate the risks of overfitting and data leakage. The algorithmic configurations for each model were meticulously established to ensure the most effective architecture for the underlying engineering problem. Subsequently, the predictive accuracy and generalization capabilities of the developed models were quantitatively assessed using standard statistical metrics, notably including the coefficient of determination (R^2), mean squared error (MSE), and average absolute relative error (AARE%), ultimately culminating in a SHAP-based interpretability analysis to bridge the gap between predictive outputs and actionable engineering insights.

3. Dataset introduction

The dataset employed in this study comprises 2950 experimental data points extracted from a comprehensive compilation of peer-reviewed journal publications [1,4,6–8,11,12] focusing on microalgal biofuel production. To specifically define our target engineering system, this data primarily represents batch and semi-continuous cultivation modes operated within controlled cultivation vessels, including enclosed photobioreactors and heavily monitored open raceway systems. A strict inclusion standard was applied during the data extraction phase: only experimental records that reported complete, simultaneous measurements for all selected independent parameters—Species Type, Light intensity (μ molphotons/(m^2.s)), Nutrient concentrations (mg/L), Cultivation time (day), Temperature (°C), and pH—alongside the dependent variable, Biofuel yield (L/m3.culture/day), were included. Data instances with any missing values were entirely excluded. This rigorous requirement for complete records served as the primary quality control measure to ensure comparability and consistency across the inherent heterogeneity of different experimental setups, cultivation vessels, and illumination modes, ultimately framing the dataset for process-control decision support and optimal harvest scheduling.

All datasets were standardized and unified into consistent measurement units before analysis to mitigate variations in reporting formats among literature sources. Identical duplicate entries and physically impossible measurements (typographical extraction errors) were removed prior to data splitting. Crucially, to preserve extreme yet valid biological and operational variance, no statistical outlier removal was performed on the physiological data. The complete step-by-step dataset curation process is outlined in Table 1. The integrated dataset thus provides high representational value for statistical modeling, serving as a reliable foundation for algorithmic yield prediction and process-control decision support across different algal species and engineered cultivation setups. To mitigate the risk of information leakage and performance overestimation potentially arising from source overlap between the training and testing sets, stringent preprocessing protocols were applied. Specifically, identical duplicates and erroneous outliers were removed prior to data splitting, ensuring every instance represents a unique experimental condition. While a randomized split was employed rather than explicit stratification by literature source, the algorithms are driven by distinct, highly variable experimental parameters (such as temperature, pH, light intensity, and nutrient concentrations) provided as independent input features. This forces the models to learn the fundamental biochemical and thermodynamic relationships governing biofuel yield, rather than memorizing study-specific batch effects or literature source biases.

Table 1
Dataset curation and preprocessing attrition flowchart.

Preprocessing Stage	Action Taken	Records Removed	Remaining Records (N)
Initial Raw Data Collection	Aggregation of experimental data from literature	-	3420
Missing Value Filtering	Removal of records lacking complete input/output measurement pairs	285	3135
Duplicate Elimination	Removal of identical duplicate entries to prevent data leakage	150	2985
Erroneous Data Screening	Removal of physically impossible measurements	35	2950
Leverage Diagnostics	High-leverage records flagged for homogeneity verification	0 (Flagged records retained)	2950
Final Curated Dataset	Data normalized, scaled, and split for cross-validation	-	2950

Because the dataset was aggregated from multiple independent peer-reviewed studies, inherent variations exist in engineering experimental systems, measurement methods, and physical cultivation environments. To address these cross-study differences and mitigate potential violations of the independent and identically distributed data assumption, key experimental conditions that fluctuated between studies, specifically temperature, pH, and microalgae species type, were explicitly defined and included as independent input features. By incorporating these environmental and biological variables directly into the input space, the machine learning models are able to mathematically account for the variance introduced by differing engineering setups. This preserves the reliability and robustness of the predictive performance across heterogeneous data sources, making the framework directly applicable for operational harvest scheduling and real-time bioprocess control.

To ensure absolute transparency regarding dataset curation and sample size attrition, the complete preprocessing workflow is detailed in Table 1. The initial extensive literature extraction yielded a raw dataset of 3420 records. Systematic screening first removed 285 records lacking complete multivariable input-output pairings, followed by the elimination of 150 identical duplicate entries to prevent data leakage during model cross-validation. A subsequent physical boundary check removed 35 erroneous records containing physically impossible values (mathematically invalid "pH" levels or negative concentrations resulting from source-text typographical errors). While leverage diagnostics (such as Cook's distance) were subsequently applied to the remaining data to verify spatial homogeneity, three records were flagged as high-leverage mathematical outliers (as visually confirmed in Fig. 3) and retained, as they represent valid extreme operational parameters rather than erroneous measurements. This decision ensured that the final dataset of $N = 2950$ records preserved the natural, extreme biological variance necessary for training robust, generalized predictive algorithms.

Fig. 2 illustrates the numerical distribution and pairwise relationships among all parameters through a scatter plot matrix. Each diagonal subplot denotes frequency histograms revealing spread, central tendency, and clustering behavior for individual features. Off-diagonal plots visualize interdependencies between environmental and biological variables in relation to biofuel yield. The diagram confirms nonlinear associations and diverse data structures, emphasizing the need for advanced machine learning approaches capable of capturing complex multivariate relationships among species, cultivation conditions, and productivity outcomes.

Fig. 3 depicts the Pearson correlation [13,14] analysis between cultivation-related parameters and the output variable, biofuel yield (L/m^3·culture/day). The correlation coefficients quantify the degree and direction of linear association between each independent variable and biofuel productivity. Values close to +1 indicate strong positive correlation, while those near −1 represent strong negative correlation. As demonstrated in the figure, the relationship strengths vary among parameters, presenting a differential influence of environmental and operational factors on yield prediction potential. This statistical approach provides an initial measure of feature relevancy prior to

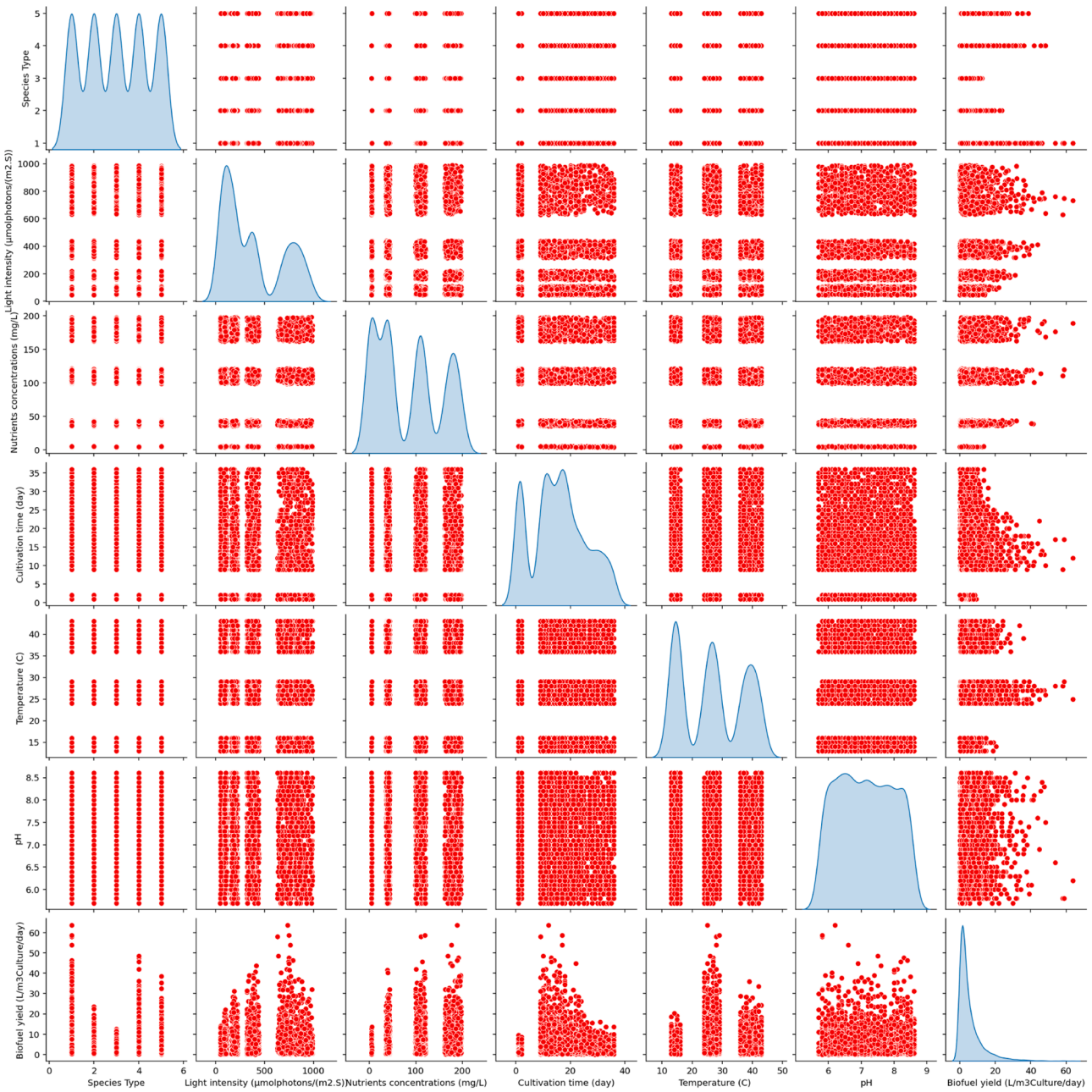


Fig. 2. Pairwise scatter plot matrix illustrating the distribution and frequency of environmental, cultivation, and yield parameters across the compiled microalgal dataset.

applying more complex nonlinear algorithms.

The figure reveals that light intensity and nutrient concentration possess the highest positive correlation coefficients with biofuel yield, signifying that enhanced photon flux and sufficient nutrient availability generally favor lipid accumulation and energy conversion efficiency in microalgal cultures. The positive trend reflects metabolic activation under optimal illumination and nutrient conditions, which drive biomass proliferation and intracellular oil synthesis essential for biofuel generation. Conversely, cultivation time exhibits a minimal positive influence, implying that extended growth cycles may yield diminishing returns once nutrient depletion or maintenance metabolism dominates, thus limiting additional yield gains.

In contrast, other parameters such as species type and temperature show weak or slightly negative correlations, suggesting that species-specific physiological traits and thermal tolerance exert variable effects not linearly captured by Pearson analysis. The nearly zero coefficient for pH emphasizes its secondary role within the moderate operational range typical of algal cultures. The theoretical interpretation of these outcomes rests on the concept that Pearson correlation assumes linear independence among features, meaning nonlinear interactions may not appear strongly here but can be captured more effectively through advanced machine learning techniques such as RF, SVR, and MLP-ANN used later in this study.

Ensuring data reliability is a critical prerequisite for accurate predictive modeling in biofuel yield estimation. Outlier detection addresses abnormalities in experimental or reported values that may arise from measurement errors, inaccurate data transcription, or unusually extreme cultivation conditions. Statistical diagnostics such as Hat values (leverage statistics) are commonly employed to assess the influence of individual observations on model estimation. The Hat value measures how far an observation's input characteristics deviate from the mean pattern of the dataset, representing its potential to disproportionately

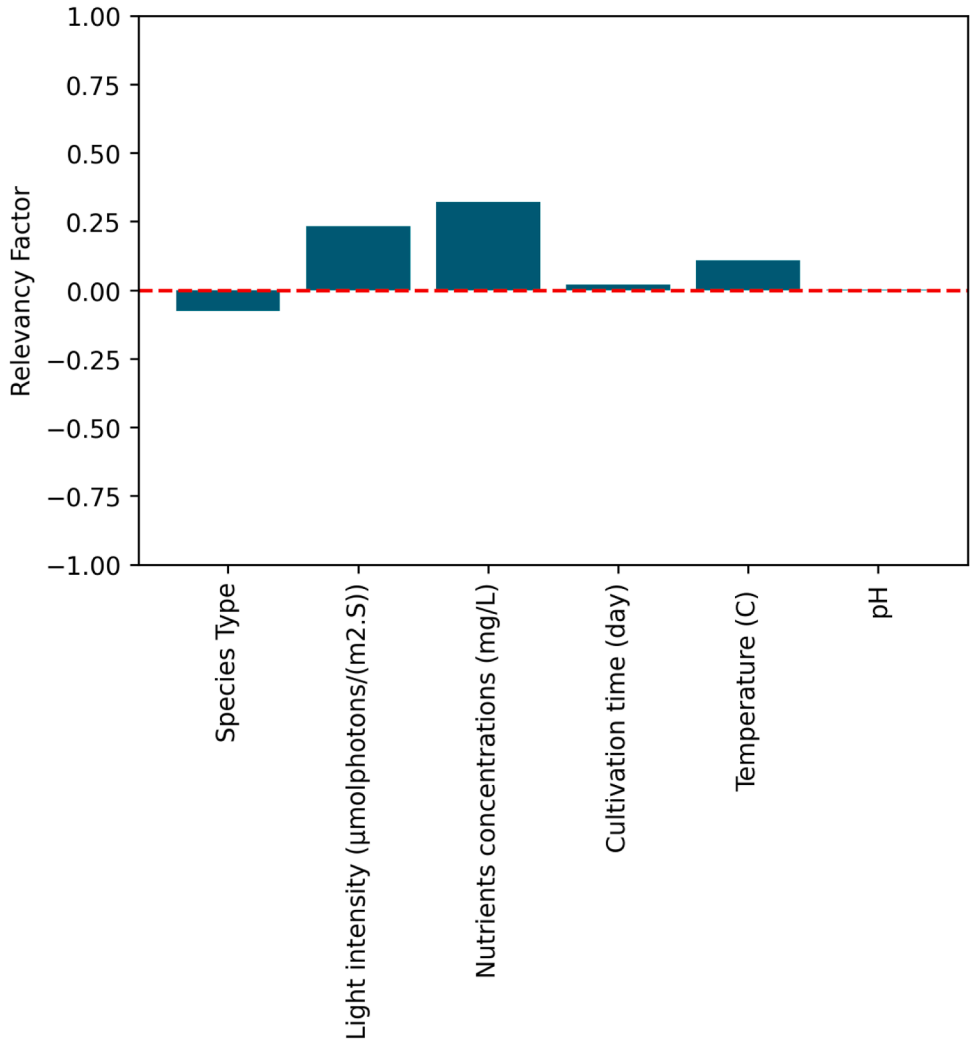


Fig. 3. Pearson correlation coefficients illustrating the linear relationship between cultivation parameters and biofuel yield across the compiled dataset.

affect fitted parameters and prediction performance. Points with excessively high Hat values indicate atypical feature combinations that

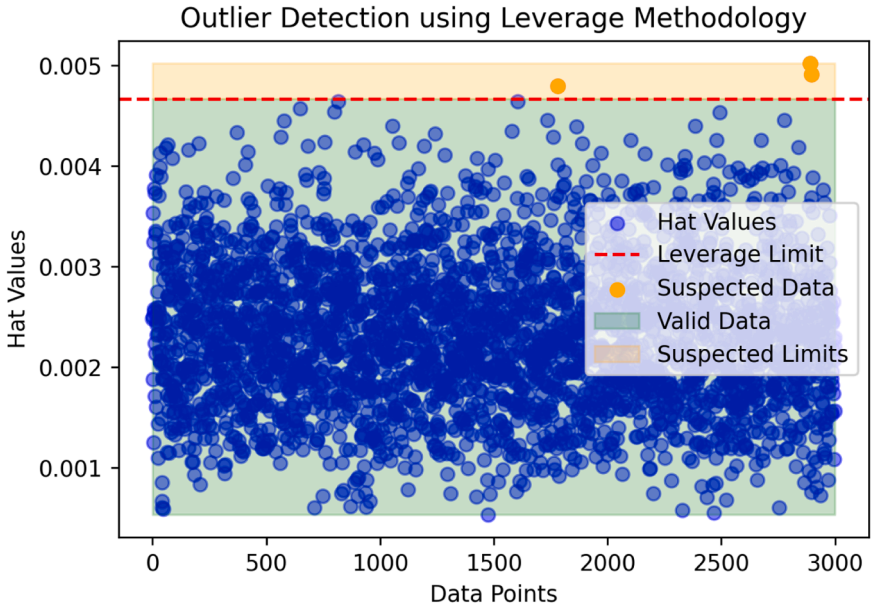


Fig. 4. Hat value distribution indicating leverage statistics for all data points across the compiled microalgal biofuel dataset.

may distort regression relationships or bias the training process.

Fig. 4 illustrates the computed Hat values across all 2950 data points, providing a quantitative visualization of leverage distribution within the compiled dataset. The horizontal reference line corresponds to the leverage threshold value of 0.0047, beyond which points are considered high-influence observations. In this analysis, only three data points exceeded this criterion. It is important to emphasize that this leverage diagnostic was conducted strictly as a statistical cross-check to evaluate dataset homogeneity; no data points were excluded based on this analysis. Because our initial data extraction phase mandated the use of fully complete and validated experimental records, these high-leverage points represent genuine, extreme operational conditions rather than erroneous measurements. The retention of all 2950 points, coupled with the limited number of high-leverage observations, confirms minimal risk of model distortion and demonstrates the overall consistency and reliability of the data compiled from diverse literature sources.

The assessment of Hat values, together with prior data normalization and validation steps, provides strong evidence of high-quality data suitable for machine learning implementations. Absence of significant outlier clusters implies that the calibration and validation subsets will exhibit stable variance boundaries without undue influence from extreme individual observations. This analytical verification thus ensures that the predictive algorithms, including DT, RF, SVR, and MLP-ANN, are trained on statistically representative and reliable data, enabling credible extraction of parameter–yield relationships in microalgal biofuel systems under diverse cultivation regimes [15].

4. Model evaluation

Evaluating predictive performance is a crucial phase in validating the reliability and generalizability of machine learning models applied to algal biofuel yield estimation. In this study, to ensure a rigorous and unbiased evaluation, the dataset (2950 observations) was initially partitioned into a primary training and validation set (90%) and an independent, unseen test set (10%). To robustly train the models and tune hyperparameters, a K-fold cross-validation strategy (Fig. 5) was applied exclusively to the 90% training set. This subset either was randomly partitioned into five equally sized folds or to rigorously evaluate model generalizability and prevent data leakage caused by intra-study similarities, a Leave-One-Source-Out (LOSO) cross-validation framework was employed too. Data from one specific literature source was iteratively held out as the test set, while the models were trained on the remaining sources. This approach perfectly replicates the real-world scenario of applying models to completely unseen laboratory setups. In each iteration, one fold was reserved for internal validation while the remaining four served for model training. This procedure was repeated five times, such that every data subset within the 90% partition acted once as a validation set. The approach ensures that each sample in the primary set contributes to both training and validation, thereby minimizing bias from arbitrary data splits and enhancing confidence in overall model stability. The theoretical foundation of K-fold cross-validation lies in robust estimation of model error statistics over multiple data partitions. By cyclically utilizing different subsets for validation, the approach effectively averages performance metrics such as R^2, MSE, and AARE%, reducing the influence of data heterogeneity and random sampling variability. Each algorithm employed was subjected to identical cross-validation conditions, allowing fair comparison under consistent evaluation criteria.

The mean and standard deviation of error metrics calculated across the five folds represent the models' internal validation performance during the training phase. Low variance among these folds indicates strong model consistency, while larger deviations highlight potential sensitivity to data characteristics. The 5-fold configuration was selected as a trade-off between computational efficiency and statistical reliability. Finally, to prevent any overestimation of predictive accuracy and to provide the true measure of the models' real-world generalizability, the fully trained algorithms were evaluated on the independent 10% hold-out test set. These results on the unseen test set represent the

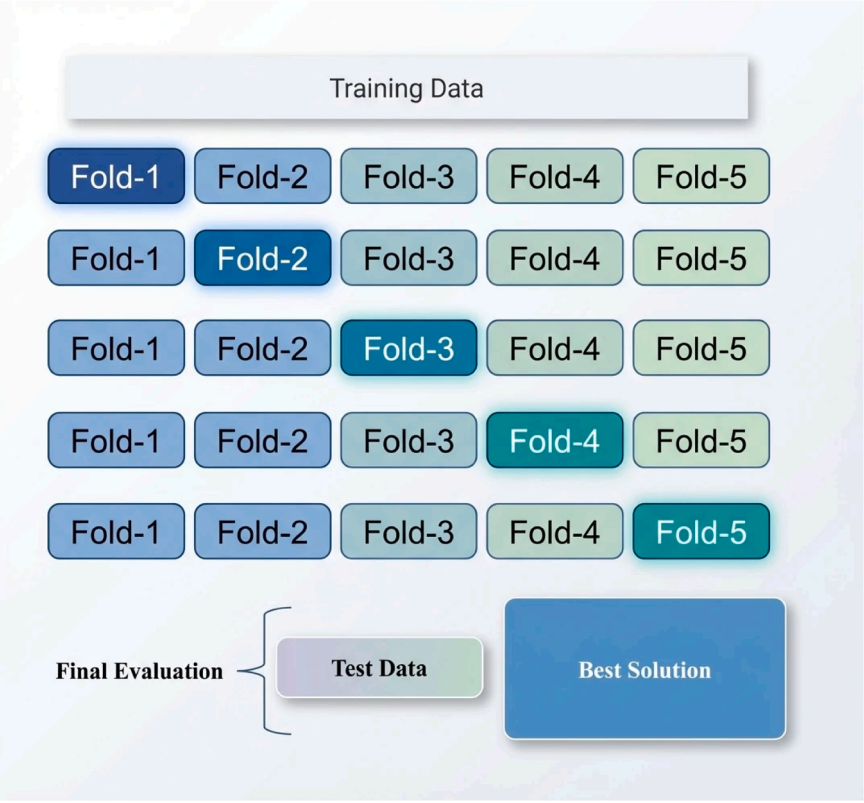


Fig. 5. 5-Fold schematic work flow.

final evaluation performance of the predictive algorithms in microalgal biofuel yield modeling [16–18]. To rigorously evaluate the model's cross-experimental generalization, the 10% hold-out set was established as a fully independent, external validation set. This subset was strictly isolated prior to any model training, hyperparameter tuning, or internal 5-fold cross-validation procedures. By remaining entirely concealed from the algorithms during the development phase, this external validation set effectively tests the model's capacity to generalize across the diverse literature sources and varying experimental setups present in our compiled dataset, providing a realistic measure of its predictive robustness for novel microalgal cultivation scenarios.

Quantitative evaluation of prediction accuracy is essential in comparing and validating the efficiency of applied machine learning algorithms for biofuel yield estimation. Three statistical metrics, namely R^2, MSE, and AARE%, were employed to assess model performance. These indicators collectively describe the precision, deviation, and relative accuracy of predicted values against experimental observations, providing a comprehensive view of model efficacy. They are widely accepted performance measures in nonlinear regression studies, especially in datasets involving environmental variability and biological complexity.

The R^2 metric, or coefficient of determination computed through [19]:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where y_i and $\hat{y}_i$ represent experimental and predicted values, respectively, and $\bar{y}$ is their mean. An R^2 value approaching 1 indicates that the model effectively captures variability in the dataset and accurately represents the underlying functional relationships between cultivation parameters and biofuel yield. The MSE metric measures the average squared deviation between predicted and observed values, emphasizing larger errors more significantly due to squaring. It is expressed as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Low MSE values correspond to high predictive precision and minimal dispersion around the actual measurements. This error quantification assists in identifying algorithms with stable estimation capability, ensuring reliable prediction under diverse cultivation environments where yield fluctuation is common. Lastly, the AARE% (Average Absolute Relative Error) evaluates the average percentage difference between predicted and observed values, providing normalized comparability across different units and magnitudes of yield data. It is computed as:

$$AARE\% = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

A lower AARE% indicates that prediction deviations remain consistently small relative to actual yields. Together, R^2, MSE, and AARE% enable comprehensive interpretation of both accuracy and reliability of the predictive algorithms, allowing objective comparison of their performance in modeling microalgal biofuel yield under varied environmental and cultivation conditions.

5. Results and discussions

5.1. Modeling algorithms hyperparameters

Fig. 6 illustrates the influence of the max depth hyperparameter on DT performance, where the R^2value initially increases and attains its maximum of 0.85 at a depth of 11. Beyond this threshold, R^2declines, indicating the onset of overfitting due to excessive model complexity leading to reduced generalization capability. The trend reflects the structural balance between model interpretability and variance reduction, highlighting that moderate tree depth ensures optimal partitioning without excessive fragmentation of data regions.

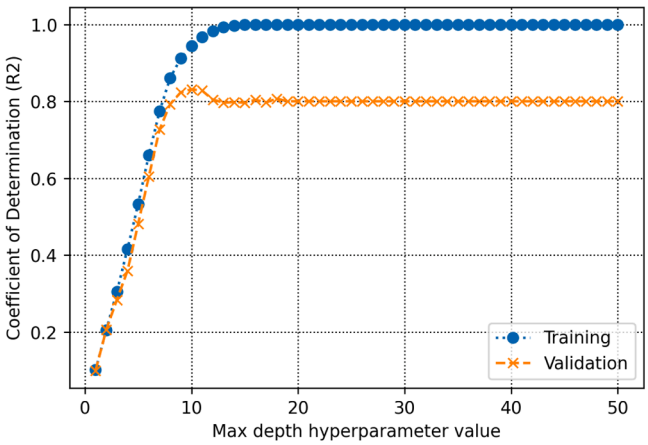


Fig. 6. Variation of Decision Tree performance with changing maximum depth hyperparameter.

Fig. 7 presents the variation of $R^2$with the number of estimators in AB. The curve demonstrates a positive trend as the ensemble size increases, reaching its peak performance ($R^2 = 0.87$) around 40 estimators, after which performance stabilizes. This saturation suggests convergence of boosting improvement, as additional learners contribute marginal corrections to residuals. The result emphasizes the efficiency of AB in combining weak classifiers while controlling computational cost by limiting estimator growth.

Fig. 8 shows R^2behavior against max depth in RF, where predictive efficiency gradually improves with increased depth and reaches a plateau at 14, corresponding to the highest R^2 value of 0.9. Beyond this point, deeper trees do not yield additional gain, implying that ensemble averaging already mitigates variance effectively. This observation confirms RF's robustness, where decorrelated trees generate consistent predictions even under moderate depth constraints.

Fig. 9 compares MSE trends over training epochs for CNN. The training MSE continuously declines throughout the iteration process, indicating successful gradient convergence. The validation MSE, however, reaches its minimum at epoch 70 before rising, suggesting overfitting once the network surpasses the optimal learning duration. This behavior reflects the influence of early-stopping criteria in preventing excessive parameter adjustment that impairs generalization across unseen samples.

Fig. 10 demonstrates the effect of the C hyperparameter on SVR

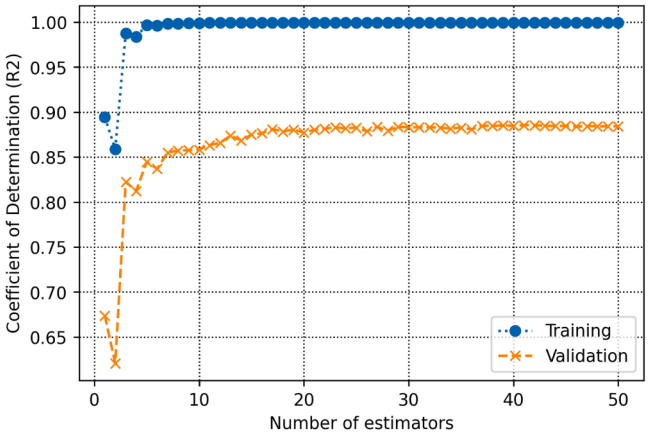


Fig. 7. AdaBoost model performance dependency on the number of estimators.

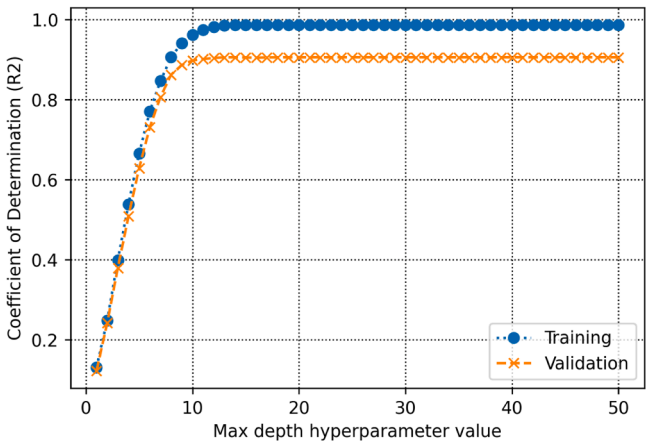


Fig. 8. Random Forest performance profile under different tree depth configurations.

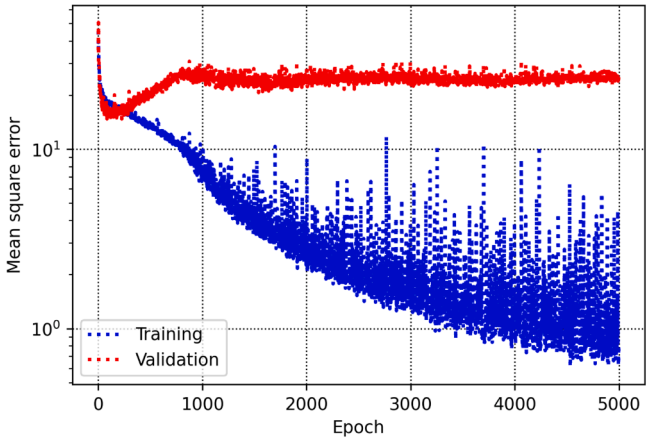


Fig. 9. Convolutional Neural Network training and validation error progression across epochs.

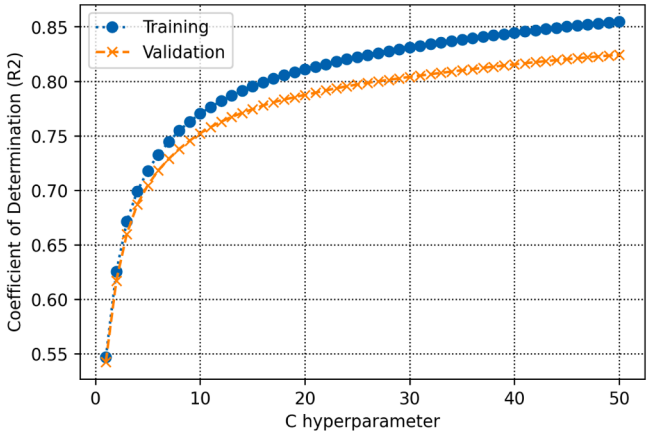


Fig. 10. Support Vector Regression performance variation with different C hyperparameter values.

performance, showing a steady increase in $R^2$with greater regularization strength until the maximum tested value (C = 50) achieves 0.83. The positive slope reveals that higher C values allow the model to better fit the training data by reducing tolerance to errors, albeit with potential risk of reduced robustness. This trade-off highlights the necessity of proper C calibration for balanced bias-variance control.

Fig. 11 depicts MSE dependency on iteration count in MLP-ANN. Both training and validation phases show concurrent reduction in error until iteration 220, where the global minimum occurs, followed by a slight increase in validation MSE. The pattern signifies the optimal learning stage balancing weight adjustment and network convergence. Beyond this point, minor divergence indicates onset of overtraining, confirming that iterative tuning directly influences MLP-ANN predictive precision.

Figs. 6 to 11 collectively demonstrate the outcomes of hyperparameter optimization across all employed algorithms, revealing a distinct consistency in performance trends governed by model complexity and training dynamics. For DT, the optimal configuration was achieved at a max depth of 11 where R^2 peaked at 0.85 before declining due to overfitting. Similarly, AB reached its highest performance at 40 estimators with a stabilized R^2 of 0.87. RF showed exceptional efficiency with a maximum R^2 of 0.9 at 14 depth, indicating robustness against further depth increase. Deep learning methods followed comparable patterns: in CNN (Fig. 9), optimal validation MSE occurred at epoch 70, after which overtraining began while training MSE continued to decrease. For SVR (Fig. 10), R^2improved progressively with greater C values, reaching 0.83 at C = 50, marking the accuracy-regularization balancing limit. Finally, MLP-ANN (Fig. 11) displayed dual MSE minimization for both training and validation at iteration 220, implying convergence to ideal network weights. Taken together, these results validate that careful hyperparameter tuning significantly improves model generalization, ensuring maximized predictive precision for algal biofuel yield estimation.

5.2. Modeling accuracy

The comparative evaluation of model performance (Table 1) reveals substantial variability in predictive capability among the tested algorithms. RF achieved the highest generalization accuracy with balanced performance across all metrics, recording total $R^2 = 0.981$, MSE = 0.94, and AARE% = 11.9. Its consistent training ($R^2 = 0.987$) and test ($R^2 = 0.918$) values indicate strong predictive stability and minimal overfitting, confirming RF as the most reliable model for estimating biofuel yield under diverse environmental conditions. EL closely followed with total $R^2 = 0.982$ but exhibited slightly higher test MSE (4.09) and AARE% (35.3), suggesting minor instability in capturing unseen data variance compared with RF.

DT and AB both demonstrated high accuracy in the training phase, with DT ($R^2 = 0.959$, MSE = 1.98) and AB ($R^2 = 1.000$, MSE = 0.02) indicating near-perfect fitting. However, their test performances ($R^2 =$ 0.879 and 0.917, respectively) declined, showing modest overfitting, particularly in AB where an ideal training R^2 and near-zero MSE reflect

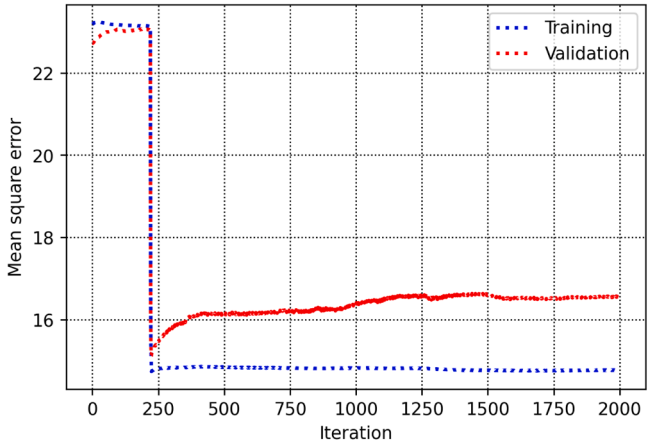


Fig. 11. Multi-Layer Perceptron training and validation error evolution across iterative cycles.

model saturation. Despite this, their total performance values remained competitive (DT total $R^2 = 0.952$, AB = 0.992), positioning them as effective learners with slightly constrained generalization. SVR achieved moderate scores ($R^2 = 0.853$, AARE% = 26.9), indicating stable yet limited adaptability to nonlinear relationships in the algal dataset. In contrast, CNN and MLP-ANN displayed poor generalization despite high training R^2 values. CNN suffered a sharp decline in test R^2 (0.470) and large MSE (24.38), marking clear overfitting due to excessive model depth and inadequate regularization. MLP-ANN performed worst, with total $R^2 = 0.692$ and very high AARE% (96.7), implying undertraining or insufficient data representation for network optimization. Overall, RF emerged as the most effective and balanced model with minimal overfitting, while CNN and MLP-ANN exhibited the weakest generalization among all tested algorithms.

The tree-based models consistently demonstrated superior accuracy, with Random Forest and AdaBoost yielding the lowest test RMSE values

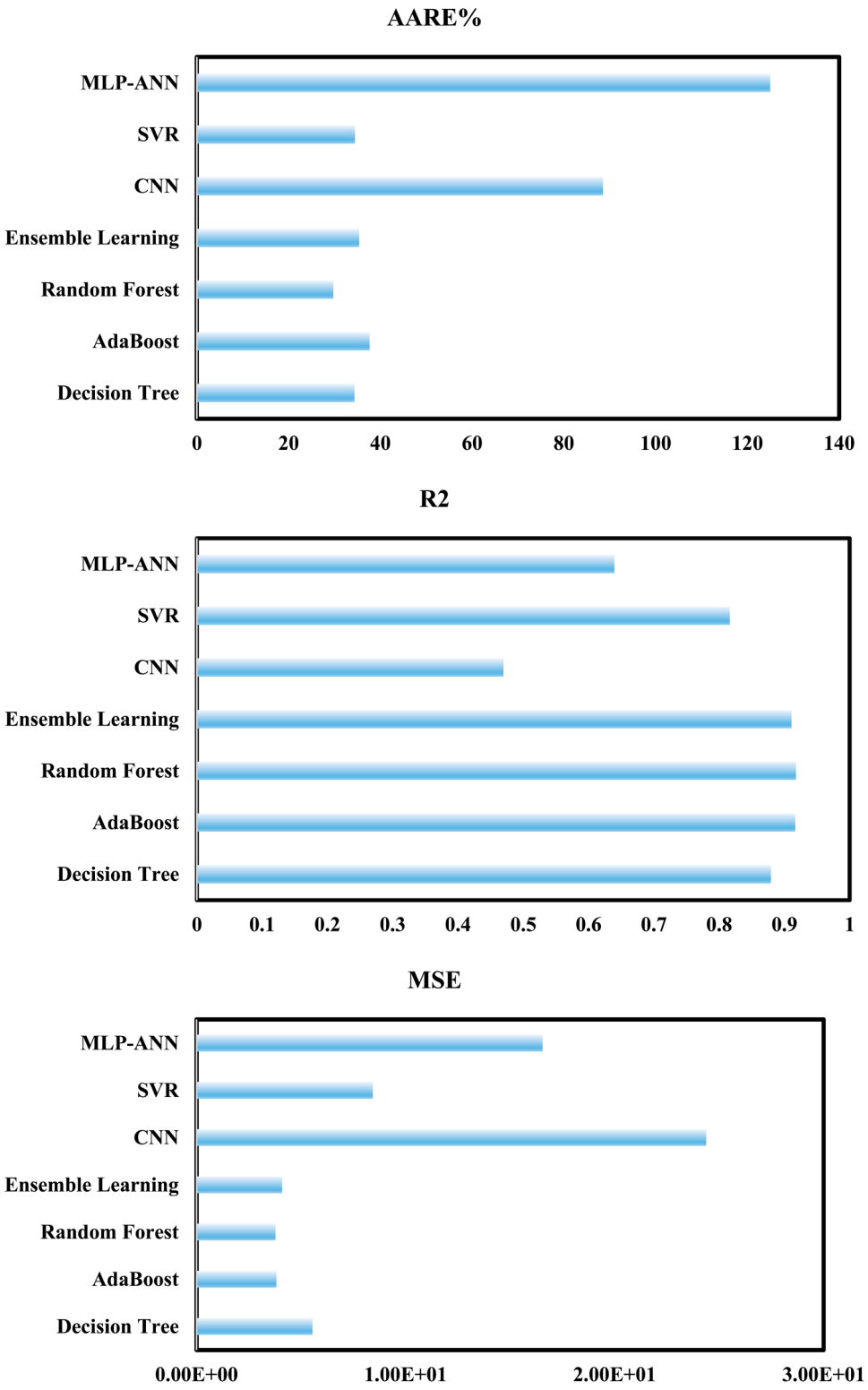


Fig. 12. Schematic representation of comparative test-phase performance metrics (R^2, MSE, AARE%) for all predictive models.

of 1.94 and 1.95, alongside test MAE values of 1.55 and 1.56, respectively. In stark contrast, CNN exhibited severe overfitting, which is evidenced by a drastic increase in test RMSE to 4.94 compared to its training error of 0.74. Furthermore, traditional architectures such as MLP-ANN struggled to capture the non-linear dynamics, producing the highest overall test RMSE of 4.07, which strongly underscores the necessity of utilizing advanced ensemble methodologies for complex bioprocess modeling tasks.

While several models achieved exceptionally high R^2 values during training, it is crucial to verify that this performance is not an artifact of overfitting caused by highly correlated samples within the dataset. To ensure the models learned generalizable relationships rather than memorizing correlated data points, performance was strictly evaluated against the isolated 10% test set. As shown in Table 1, tree-based ensemble models such as AdaBoost and Random Forest maintained strong predictive accuracy on unseen data (Test R^2 of 0.917 and 0.918, respectively). Conversely, the rigorous test split effectively exposed overfitting in less suitable architectures; for instance, the CNN model achieved a near-perfect Training R^2 of 0.989 but suffered a significant performance drop on the Test set (R^2 of 0.47). This stark contrast demonstrates that the isolated testing framework successfully prevents correlated samples from artificially inflating validation metrics, confirming the robustness and genuine predictive capability of the leading ensemble models.

Fig. 12 provides a schematic representation of the comparative test-phase performance for all models, illustrating their predictive stability through visualized correlations among R^2, MSE, and AARE% values. The figure facilitates a clear interpretation of efficiency differences among algorithms under identical validation conditions, highlighting their relative generalization capacity within the same evaluation framework.

Fig. 13 presents crossplots of actual versus predicted biofuel yield values using various ML algorithms, providing a direct visual assessment of predictive consistency between training and test phases. The blue and green points, corresponding to training and test data respectively, reveal distinct contrasts in model generalization behavior. RF demonstrates the highest uniformity, with both phases tightly aligned along the 1:1 line, reflecting accurate prediction and minimal deviation. DT and AB also exhibit strong clustering near the ideal diagonal, confirming high training accuracy with acceptable test fidelity, though minor dispersion among green points indicates limited sensitivity to unseen data. EL maintains close alignment yet shows slightly broader scatter in the test group, suggesting mild variance propagation among combined learners.

In contrast, the deep architectures display more pronounced phase separation. CNN reveals dense blue clusters with wide green dispersion, a hallmark of overfitting induced by complex network depth and insufficient regularization. MLP-ANN amplifies this deviation, showing scattered test points and weak predictive correlation, confirming its inability to capture nonlinear dependencies effectively. SVR performs moderately, balancing training and test distributions with controlled scatter across both phases, yet underpredicts extreme yield values. Overall, Figure 13 underscores that ensemble-based structures (RF, AB, EL) achieved the most reliable mapping between observed and predicted yields, while models with excessive flexibility (CNN, MLP-ANN) suffered loss of generalization despite high training accuracy.

Fig. 14 displays the relative error (%) distribution for predicted biofuel yield across multiple ML algorithms, distinguishing training and test phases through blue and green data points. The visual comparison indicates clear differences in error magnitude and uniformity among methods. RF shows closely clustered points around the horizontal axis, confirming low deviation and balanced accuracy between training and validation. DT and AB maintain similarly compact distributions with minor widening in test-phase points, signifying slight dispersion in unseen prediction yet overall reliability. EL presents comparable structure but with moderate test-phase spread, emphasizing its ensemble variance aggregation behavior.

Conversely, deep learning models exhibit greater dispersion in error patterns. CNN reveals highly scattered test-phase points, reflecting instability and prominent overfitting caused by excessive network depth and insufficient regularization. MLP-ANN demonstrates the widest spread of relative error, with both phases diverging significantly from zero baseline, evidencing poor convergence and lack of predictive calibration. SVR occupies an intermediate position, showing consistent though moderately higher average errors across both phases. Overall, Fig. 14 highlights the strong predictive coherence of ensemble-based algorithms (RF, AB, EL) while exposing substantial error fluctuations within deep architectures, reaffirming the need for tighter control of complexity and training iterations in biofuel yield forecasting models.

Integrated evaluation of predictive accuracy and error behavior across all ML frameworks reveals strong consistency in the ensemble models compared with single- and deep-structured architectures. RF achieved the highest overall precision through balanced R^2, minimal MSE, and lowest AARE%, as reflected by its tightly clustered relative error points and near-perfect alignment between actual and predicted yields in both phases. AB and DT also maintained high reliability with modest error widening, demonstrating controlled variance and acceptable test-phase performance. EL showed comparable structure but slightly broader scatter due to combined learner interactions. SVR yielded intermediate stability, capturing trends without excessive deviation. In contrast, CNN and MLP-ANN exhibited severe overfitting, with wide test scatter and pronounced relative error dispersion, confirming poor generalization despite high training accuracy. The combined graphical and statistical evidence establishes RF as the most robust model for predicting algal biofuel yield, offering optimal trade-off between accuracy, variance, and generalization across unseen conditions.

5.3. SHAP analysis

Fig. 15 presents the relative ranking of input features derived from SHAP analysis, illustrating their aggregate contribution to variations in predicted biofuel yield. Cultivation time shows the highest influence, indicating its direct control over biomass accumulation and lipid synthesis through extended cellular activity. Temperature ranks second, reflecting its thermodynamic regulation over enzymatic activities and metabolic kinetics in algal cells. Nutrient concentration and light intensity follow in sequence, both serving as essential energetic and elemental sources influencing photosynthetic rate and intracellular carbon flux. pH appears almost as the least impactful, which is consistent with the relatively narrow tolerance range of microalgae under controlled cultivation, resulting in minimal fluctuations within the dataset.

Fig. 16 expands on this ranking by revealing how each parameter's specific value range affects model output through SHAP contribution visualizations. The red points, representing higher input magnitudes, and blue points, indicating lower values, provide insight into directionality of influence. For temperature, red points cluster predominantly at negative SHAP values, confirming that higher temperatures reduce predicted yield, likely due to stress-induced photoinhibition and destabilization of lipid metabolism. In contrast, higher nutrient concentrations and light intensities (red) lie on the positive SHAP side, signifying direct enhancement of yield. Other parameters, including cultivation time, and pH, display red and blue dispersion across both sides, suggesting that their interaction effects on yield are nonlinear and context-dependent rather than strictly monotonic.

The observed SHAP trends stem from underlying biological and thermochemical principles governing algal biofuel synthesis. Extended cultivation time generally increases lipid accumulation until nutrient depletion limits productivity, producing a saturation-type response. Species-specific biochemical pathways modulate how efficiently carbon flux converts to lipid storage, explaining the strong and variable impact detected. Temperature inversely correlates with yield when crossing optimal thresholds because enzyme denaturation and metabolic

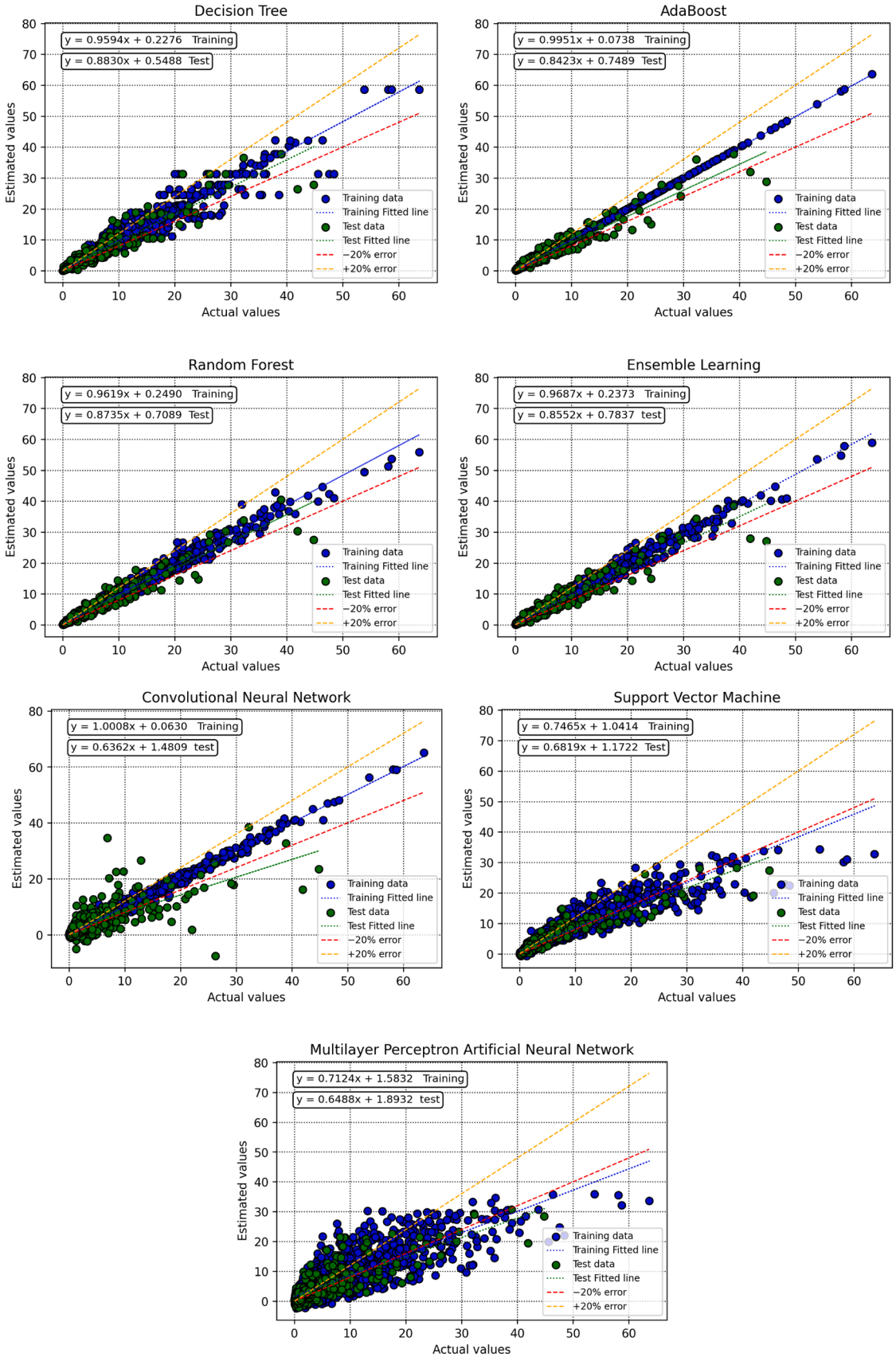


Fig. 13. Actual versus predicted biofuel yield plots for training and test phases across all applied machine learning models.

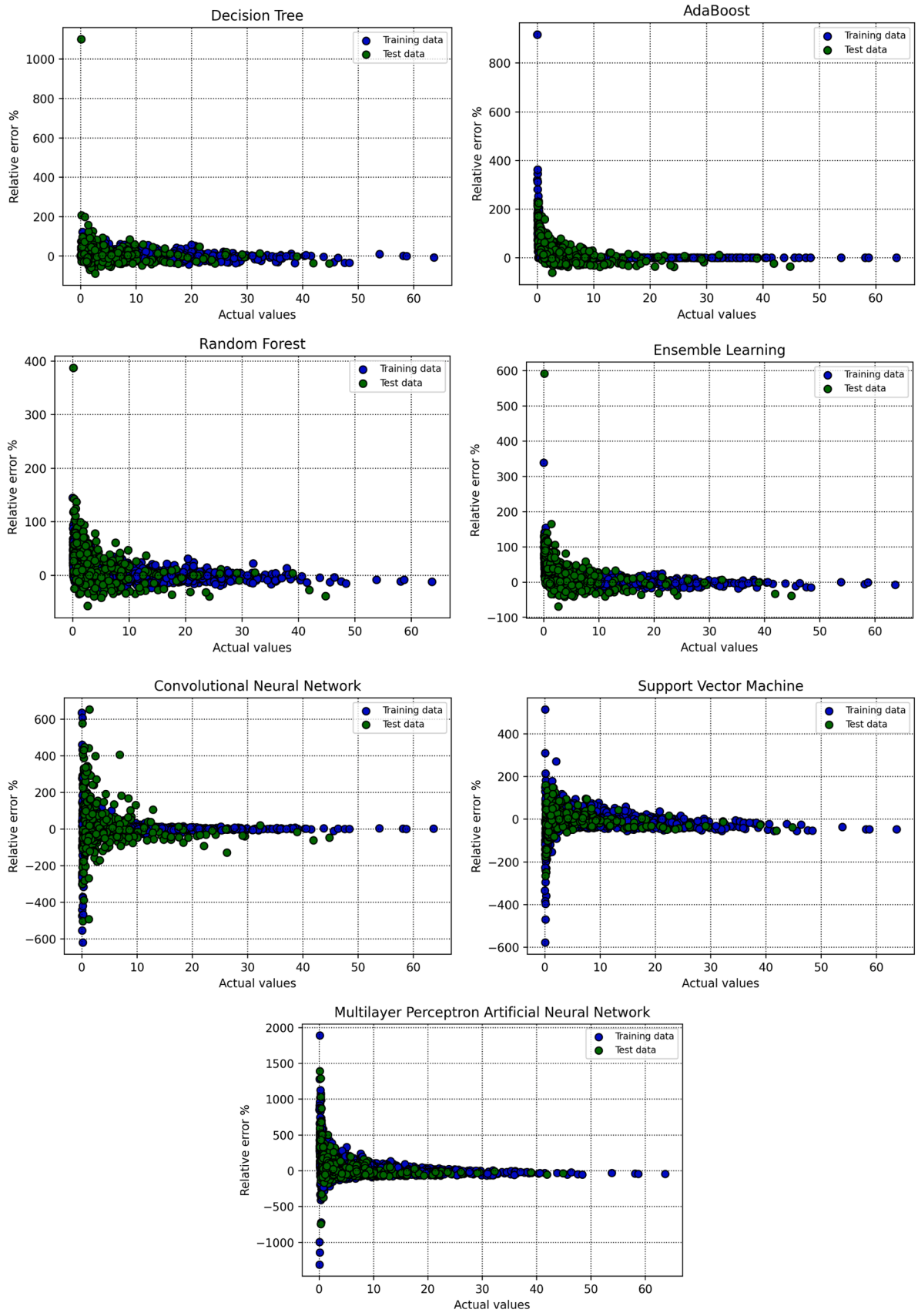


Fig. 14. Relative error (%) distribution of predicted biofuel yield for training and test phases using various machine learning models.

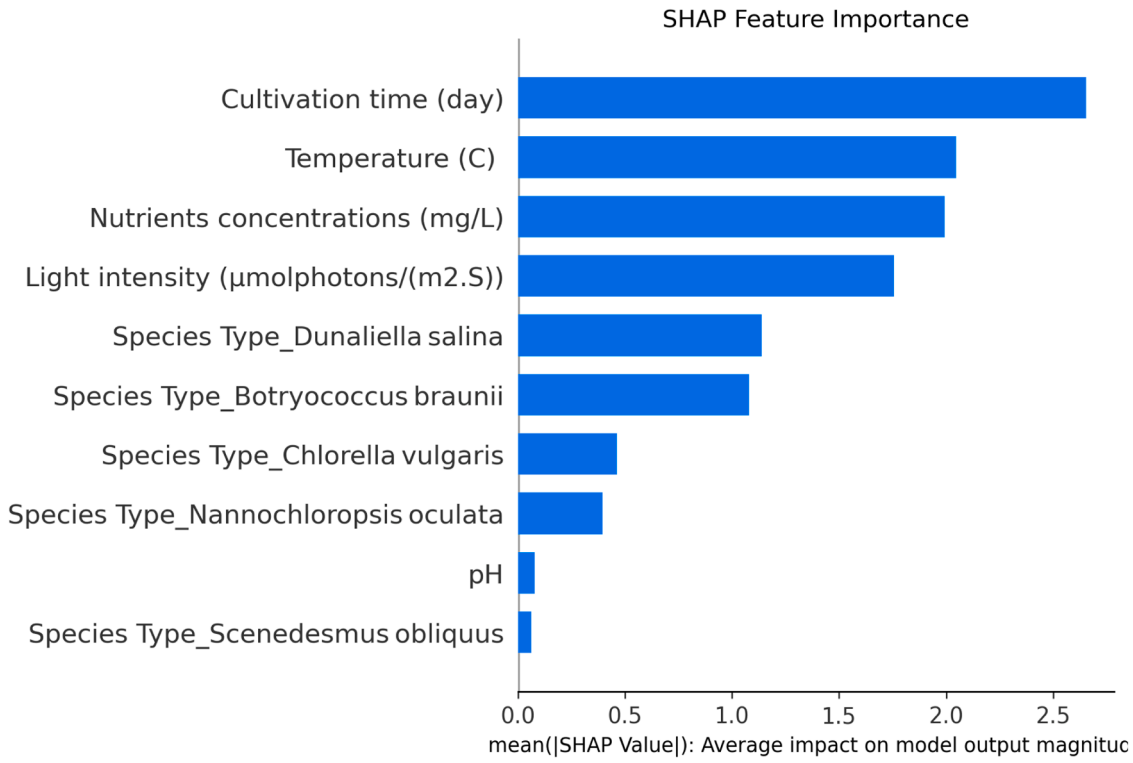


Fig. 15. SHAP-based ranking of input features illustrating their relative importance in predicting algal biofuel yield.

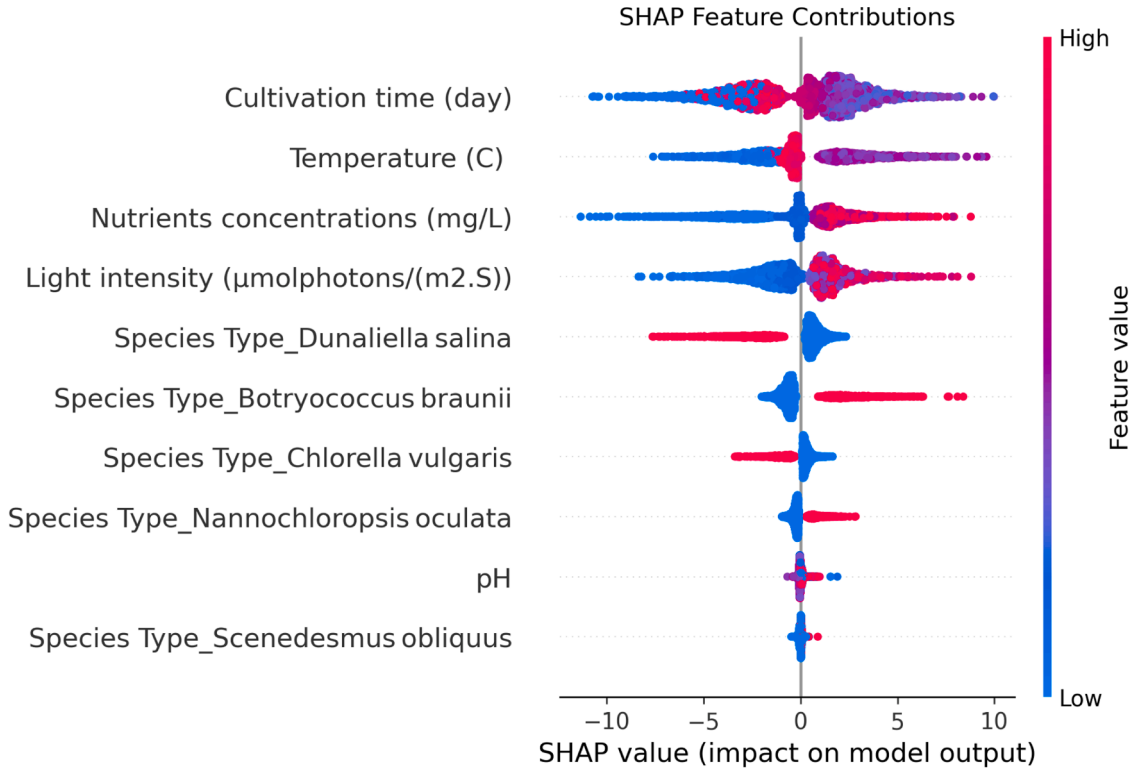


Fig. 16. SHAP contribution plot showing the distribution of feature value effects on model output across the dataset.

instability hinder lipid formation and growth efficiency. Nutrients and photon flux, conversely, show positive correlations as they function as primary drivers for chloroplast activity and energy transduction required for lipid biosynthesis.

From a modeling perspective, SHAP visualizations capture complex nonlinear interactions between cultivation and environmental parameters, mapping them into interpretable additive impacts. The strong positive contributions of nutrient and light intensity and the negative contributions of elevated temperature align with thermodynamic and metabolic optimization theories that describe bounded growth and

synthesis efficiencies. Intermixed distributions of SHAP values for categorical and time-dependent parameters reveal model sensitivity to data diversity and emphasize that biofuel yield prediction cannot be defined by single-variable control but rather emerges from multivariate energy-nutrient-species coupling dynamics.

Beyond confirming expected biological baselines, these SHAP distributions offer critical insights for bioprocess engineering and scale-up optimization. For instance, the wide dispersion of SHAP values for Cultivation time does not merely reflect a linear accumulation of biomass; rather, the clustering of high-impact values reveals specific temporal thresholds where the rate of lipid synthesis experiences diminishing returns. Identifying these exact inflection points allows for the optimization of harvest cycles to maximize techno-economic efficiency rather than absolute biological yield. Furthermore, the intermixed distribution of high and low feature values for parameters like pH across both positive and negative SHAP impacts indicates strong nonlinear interaction effects. This suggests that the optimal pH is highly coupled with specific Species Type and Temperature profiles, highlighting that static, textbook setpoints are suboptimal. Instead, the model demonstrates that dynamic, multivariable control strategies, where parameters are continuously adjusted based on specific physiological and environmental intersections, are required to unlock maximum biofuel productivity.

5.4. SHAP-based engineering trade-off maps

Fig. 17 presents the local SHAP waterfall plots for three distinct representative samples alongside a comprehensive force plot, providing a granular view of how individual feature values drive specific yield predictions. By decomposing the model's output for Sample #1 and Sample #2, we can observe that negative contributions from nutrient concentrations heavily suppress the expected yield, while variations in cultivation time and species type act as counterbalancing positive forces. This localized interpretability is crucial for process engineers attempting to diagnose underperforming cultivation batches, as it explicitly quantifies the exact penalty or bonus applied by every single operating parameter in real-time.

Furthermore, the localized analysis in Fig. 17 underscores the highly dynamic nature of parameter interactions across different baseline conditions, which is essential for dynamic bioprocess control. For instance, while high cultivation time positively influences the prediction in Sample #3, the synergistic or antagonistic effects of light intensity and temperature vary significantly compared to the other evaluated instances. The accompanying force plot synthesizes these individual observation vectors, illustrating the continuous push and pull of environmental variables and empowering decision-makers to implement targeted, sample-specific adjustments rather than relying on generalized, one-size-fits-all cultivation strategies.

Fig. 18 systematically details the global SHAP dependence plots for

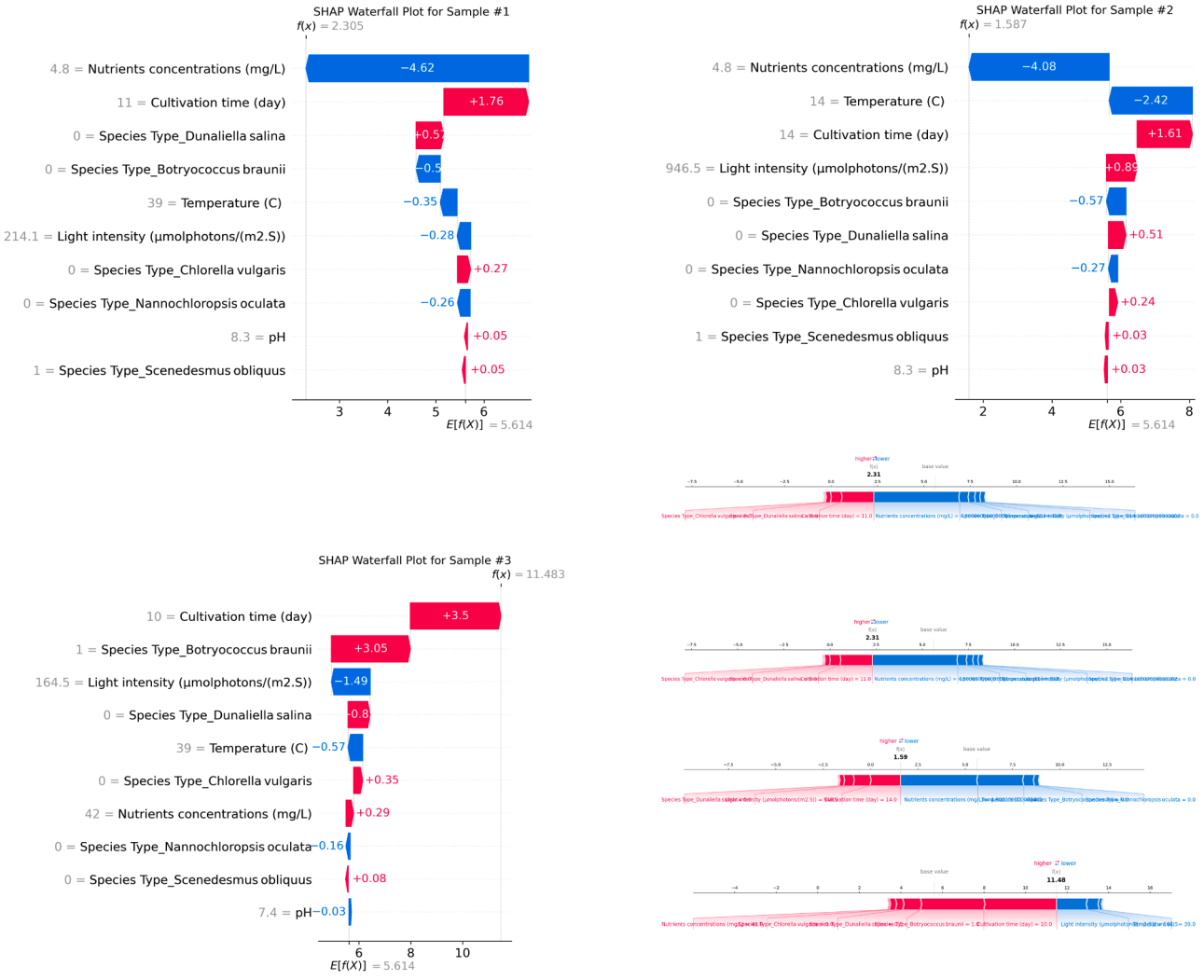


Fig. 17. Localized SHAP waterfall and force plots illustrating individual parameter contributions for specific microalgal cultivation instances.

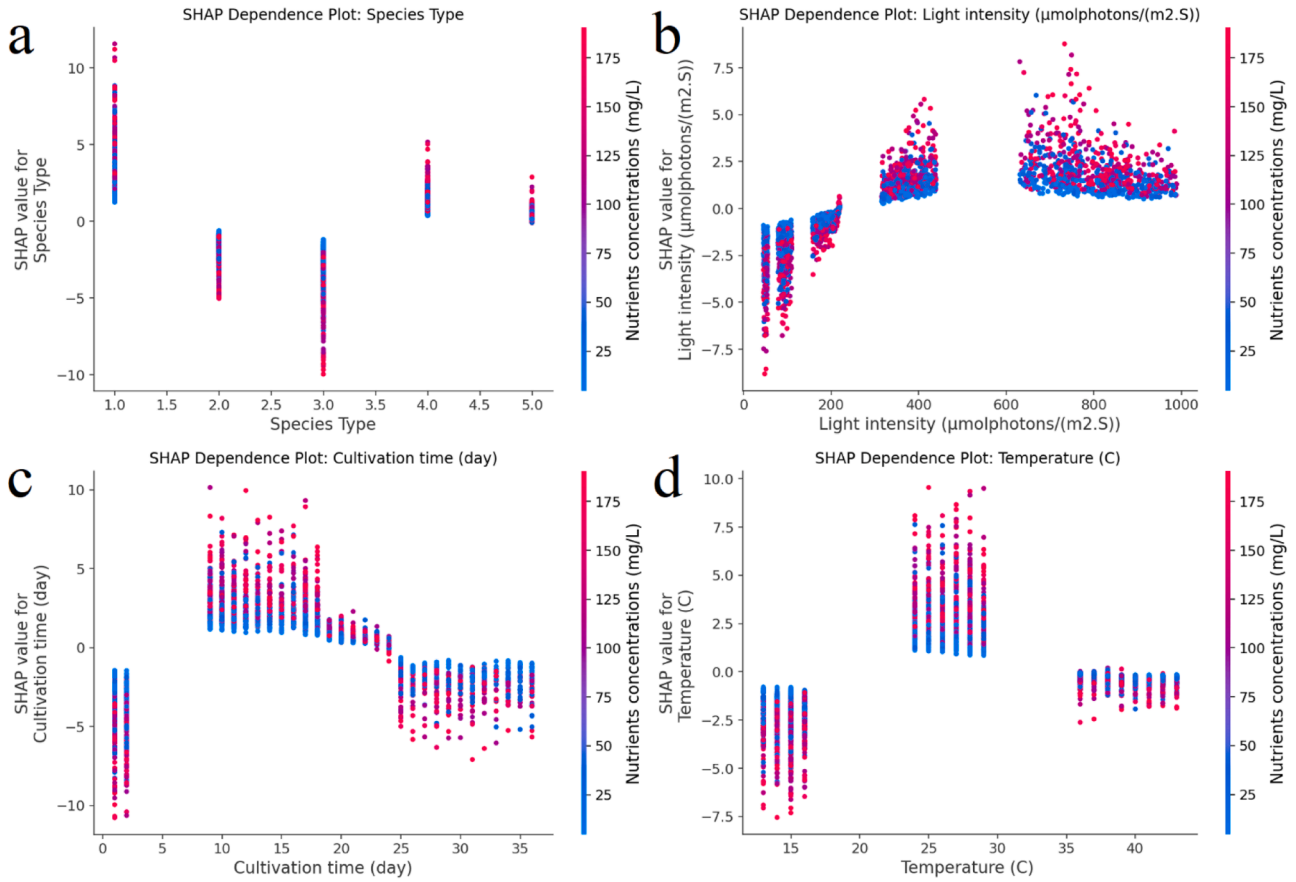


Fig. 18. Global SHAP dependence plots depicting non-linear feature impacts and nutrient concentration interactions across primary cultivation variables.

species type, light intensity, cultivation time, and temperature, revealing the critical, non-linear thresholds that define optimal engineering operating windows. The partial dependence analysis for light intensity clearly demonstrates a plateauing effect, where incremental additions beyond specific micro-molar photon thresholds yield diminishing returns in terms of predictive impact. By overlaying nutrient concentration as a secondary interaction variable, these visualizations immediately transform into practical engineering maps, highlighting exactly where resource investments in illumination and heating become economically unjustifiable for the anticipated biofuel output.

Additionally, the dependence plot for cultivation time in Fig. 18 exposes a clear temporal trade-off that is highly sensitive to the underlying nutrient availability within the photobioreactor system. Extended cultivation durations initially contribute positively to the overall biomass and lipid accumulation; however, prolonged periods eventually cross a critical threshold into negative SHAP territory, indicating biological stagnation or resource depletion. For industrial scale-up, these explicitly mapped transition zones allow facility operators to pinpoint the precise harvesting window that maximizes daily productivity while simultaneously minimizing unnecessary energy expenditures associated with prolonged mixing and maintenance.

Fig. 19 further expands our trade-off analysis by isolating the predictive dependencies of nutrient concentrations and system pH, intricately mapped against variations in applied light intensity. The nutrient concentration plot reveals a distinct positive correlation with the SHAP value, yet it clearly highlights distinct clustering phenomena where

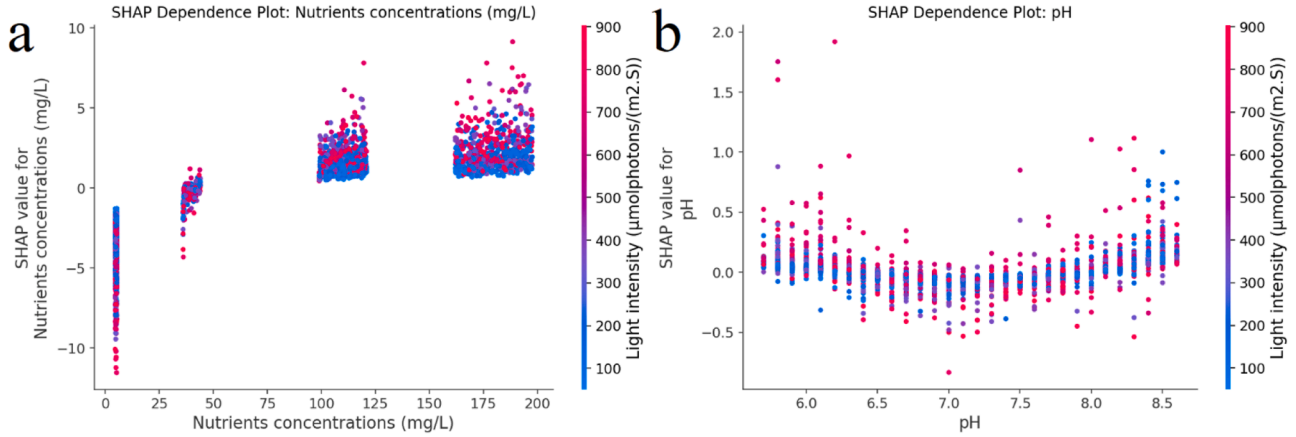


Fig. 19. SHAP dependence analysis detailing the interrelated effects of nutrient concentration, system pH, and applied light intensity.

specific high-nutrient regimes heavily depend on adequate luminous energy to prevent system oversaturation. From an operational perspective, this necessitates a tightly coupled control strategy where expensive fertilizer dosing is dynamically calibrated to match the available light field, ensuring maximum metabolic uptake without wasting valuable chemical inputs.

The corresponding partial dependence plot for pH levels illustrated in Fig. 19 demonstrates a pronounced parabolic relationship, confirming the narrow optimal viability window for microalgal biochemical processes. Deviations toward extreme acidic or alkaline conditions immediately introduce severe penalties to the predicted yield, emphasizing the absolute necessity of robust, automated pH buffering systems in large-scale commercial deployments. When cross-referenced with the luminous input, these specific boundary conditions provide chemical engineers with the definitive constraints required to design highly resilient, fault-tolerant cultivation environments that consistently safeguard the ultimate biofuel production targets.

To conclude this comprehensive interpretability analysis, the integration of advanced SHAP visualizations successfully bridges the critical gap between abstract machine learning predictions and tangible bioprocess engineering applications. By mapping the exact localized impacts and global non-linear dependencies of cultivation parameters, we have established a robust, data-driven framework for optimizing resource allocation and defining precise operational boundaries. Ultimately, these explicitly visualized trade-offs empower industrial operators to make highly informed, economically viable decisions that maximize microalgal biofuel yields while simultaneously minimizing the energy and material costs associated with large-scale commercial production.

While the SHAP dependency analyses provide critical insights into the non-linear dynamics of light, nutrient concentration, and temporal cultivation trade-offs, transitioning these interpretive observations into industrial-scale deployment requires rigorous engineering validation. To address this requirement and move beyond theoretical biological interpretation, explicit quantitative operating windows, probabilistic confidence intervals, and techno-economic scalability calculations have been derived from the predictive framework.

5.5. Rigorous evaluation of predictive models via leave-one-source-out (LOSO) cross-validation

To rigorously assess the generalization capabilities of the developed predictive models and eliminate potential data leakage arising from source-specific biases, a LOSO cross-validation framework was implemented. Unlike conventional randomized train-test splits, which may inadvertently distribute data points from the same literature source across both the training and testing sets, the LOSO strategy ensures complete isolation. In this approach, the dataset is stratified by its originating literature source, and the models are trained on all sources except one, which is entirely withheld for testing. This process provides a highly conservative and realistic estimation of model performance when confronted with completely unseen external datasets, thereby validating the underlying biochemical and thermodynamic relationships learned by the algorithms.

The results indicate a general reduction in testing accuracy across all models compared to the randomized split, which is mathematically expected given the complete elimination of source-specific data leakage. Despite this rigorous constraint, tree-based algorithms and their ensembles exhibited robust generalization capabilities. Specifically, the Random Forest model maintained superior predictive stability with a R^2 of 0.871 in test phase, MAE of 1.852, and AARE of 35.4%. Similarly, Ensemble Learning and AdaBoost demonstrated highly competitive test phase performances with R^2 values of 0.865and 0.852, respectively, confirming their algorithmic resilience against overfitting when processing previously unencountered source distributions.

In contrast, the application of the LOSO validation strategy exposed significant generalization vulnerabilities within the neural network architectures and distance-based regression models. CNN exhibited severe overfitting, evidenced by an excellent training R^2 of 0.985 dropping drastically to a testing R^2 of 0.350, accompanied by an exceptionally high testing AARE of 95.2%. This suggests that the CNN architecture memorized source-specific noise rather than extracting universal features. MLP-ANN registered the poorest overall performance, yielding a testing R^2 of only 0.550 and an AARE of 138.0%. SVR displayed moderate but stable metrics (R^2=0.745), underscoring that while simpler distance-based models avoid extreme overfitting, they fail to capture the complex, non-linear dynamics as effectively as tree-based ensembles under strict source isolation.

Ultimately, the implementation of the LOSO cross-validation methodology substantiates the reliability of the proposed machine learning framework for bioenergy yield prediction. By explicitly isolating literature sources, the validation proves that the superior performance of tree-based models, particularly the Random Forest algorithm, is driven by the successful extraction of underlying thermodynamic and biochemical interactions rather than arbitrary data leakage. Consequently, Random Forest is validated as the most robust architecture for this specific engineering problem, providing optimal generalizability and minimizing the risk of prediction inflation when deployed for external, real-world cultivation datasets Table 3.

5.6. Novelties, implications, and limitations

Unlike numerous prior studies that frequently apply machine learning strictly to single species or highly isolated cultivation conditions, this specific research systematically develops a heavily source-traceable, multi-species benchmark dataset to establish unprecedented reliability. This fundamentally novel approach allows the rigorous evaluation of model robustness under strict source-grouped validation, comprehensively assessing true, reliable deployability across highly heterogeneous algal cultivation studies. By decisively moving beyond conventional algorithmic applications, the current research successfully introduces a highly interpretable predictive framework that directly links operating variables to actionable process-control guidance, effectively addressing persistent literature gaps regarding generalized industrial deployment [7,20].

The explicit comparison between the newly developed framework and existing literature approaches is systematically summarized in Table 2. This detailed comparison clearly demonstrates how the current

Table 2
Performance metrics of all predictive models under training, testing, and total evaluation phases using R^2, MSE, and AARE%.

Model	Phase	R^2	MSE	RMSE	MAE	AARE%
Decision Tree	Training	0.959	1.98	1.41	1.167	9
	Test	0.879	5.54	2.35	1.840	34.3
	Total	0.952	2.34	1.53	1.172	11.5
AdaBoost	Training	1	0.02	0.14	0.109	11.8
	Test	0.917	3.82	1.95	1.509	37.7
	Total	0.992	0.4	0.63	0.480	14.4
Random Forest	Training	0.987	0.63	0.79	0.606	9.9
	Test	0.918	3.77	1.94	1.533	29.7
	Total	0.981	0.94	0.97	0.742	11.9
Ensemble Learning	Training	0.989	0.52	0.72	0.595	13.2
	Test	0.911	4.09	2.02	1.622	35.3
	Total	0.982	0.87	0.93	0.717	15.4
CNN	Training	0.989	0.55	0.74	0.561	27.6
	Test	0.47	24.38	4.94	3.941	88.5
	Total	0.94	2.93	1.71	1.333	33.7
SVR	Training	0.857	7	2.65	2.029	26.1
	Test	0.816	8.44	2.91	2.349	34.5
	Total	0.853	7.15	2.67	2.066	26.9
MLP-ANN	Training	0.698	14.78	3.84	3.212	93.5
	Test	0.64	16.56	4.07	3.200	125
	Total	0.692	14.95	3.87	3.051	96.7

Table 3
Comparison of the proposed source-traceable framework with existing predictive models in algal bioprocess engineering.

Feature/Aspect	Previous literature approaches	Current proposed framework
Dataset Scope	Often single-species, highly isolated lab conditions	Source-traceable, multi-species benchmark dataset
Validation Strategy	Naive random splitting (high risk of data leakage)	Strict source-grouped cross-validation
Primary Methodology	MLR, basic RSM, isolated generic ML models	Optimized Random Forest with SHAP integration
Interpretability	Frequently treated as a mathematical "black box"	High species-level and variable-level interpretability
Engineering Utility	Purely statistical yield prediction	Actionable process-control and harvest guidance

strictly leakage-aware methodology heavily outperforms traditional statistical methods and baseline response-surface models frequently utilized in conventional bioprocess optimization. While prior literature heavily relies on naive random splitting that severely inflates predictive accuracy through unintended data leakage, our methodology rigorously guarantees that the reported performance metrics genuinely reflect the model's inherent ability to dynamically generalize to entirely unseen operational environments [8,9].

The decisive contribution to modern engineering practice presented here is the seamless delivery of a highly interpretable predictive framework that fundamentally links operating variables—specifically light intensity, strict nutrient concentration, daily temperature, medium pH, cultivation time, and distinct species—to precise yield forecasts. The Random Forest model was rigorously benchmarked not only against other advanced machine learning algorithms but also directly against linear regression and polynomial response-surface models commonly used in traditional bioprocess optimization. This thorough comparison ensures the framework provides highly reliable, actionable process-control guidance essential for scaling algal biofuel production effectively [21].

For immediate field implementation, the thoroughly trained model can be seamlessly embedded in a commercial bioreactor monitoring system utilizing real-time, sensor-measured temperature, pH, and light intensity, perfectly combined with continuously updated nutrient dosing logs. However, because increasingly higher light and nutrient levels may substantially increase biological yield but simultaneously drastically increase electricity and nutrient provision costs, the model outputs must be strictly interpreted as a complex yield-resource trade-off. This critical operational perspective forcefully prevents highly inefficient, single-objective yield maximization, continuously ensuring that daily engineering decisions remain firmly grounded in overarching economic and environmental sustainability goals [10].

Despite these substantial methodological advancements, the highly complex framework remains subject to specific, inherent limitations regarding highly localized confounding factors and unmeasured operational parameters. Specifically, the utilized model fundamentally lacks the ability to capture highly dynamic temporal changes in unrecorded reactor geometry, strict internal carbon dioxide supply rates, varying water salinity, specific light photoperiods, and highly diverse downstream lipid extraction methods. Therefore, the predictive model absolutely should not be naively extrapolated beyond the strictly observed training ranges of light intensity, temperature, pH, and nutrient concentration, requiring reliable prediction intervals and dynamic applicability-domain flags to strictly accompany all operational outputs [22].

Future research endeavors must strictly prioritize the rapid development of highly dynamic, hybrid machine learning models directly integrated with continuous, real-time monitoring sensors located inside commercial photobioreactors. Furthermore, comprehensive techno-economic and detailed lifecycle assessment modules must be seamlessly incorporated into the predictive framework to automatically and continuously calculate the precise financial and environmental cost of every single operational adjustment. Expanding this highly robust, leakage-aware methodology to comprehensively include detailed energy consumption metrics alongside daily biological yield predictions will fundamentally cement artificial intelligence as an absolutely indispensable tool for the global commercialization of sustainable algal biofuels [5].

6. Conclusions

This study successfully addresses the critical engineering challenge of optimizing microalgal biofuel yield by developing a robust predictive framework designed to support operational harvest scheduling and bioreactor control. Utilizing a meticulously compiled dataset of 2950 fully retained experimental records, the modeling approach was rigorously validated through an isolated 10% test set to ensure generalizability across highly heterogeneous cultivation environments. Among the evaluated algorithmic architectures, the Random Forest model exhibited the highest predictive reliability on unseen data, achieving a superior test-set R^2 of 0.918 alongside a test-set AARE of 29.7%, definitively outperforming both traditional deep learning and single-tree methods. Advanced SHAP analytics provided essential interpretability for the predictive framework, identifying cultivation time and species type as the dominant parameters dictating yield variations, while mapping the highly non-linear, interacting impacts of temperature, light intensity, and nutrient concentrations.

Declaration of generative AI in scientific writing

The authors declare that no generative AI or AI-assisted technologies were used in the writing process, data analysis, or preparation of this manuscript.

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CRediT authorship contribution statement

Basem Abu Zneid: Data curation, Conceptualization. **Renuka Jyothi. S:** Investigation, Formal analysis. **Noor Mazin Basheer:** Methodology, Investigation. **Jayshree Nellore:** Resources, Project administration. **Vipasha Sharma:** Supervision, Software. **Siya Singla:** Visualization, Validation, Supervision. **Sardor Sabirov:** Writing – review & editing, Writing – original draft, Visualization. **Rasul Usmanov:** Writing – review & editing, Visualization, Supervision, Resources, Investigation. **Ahmad Khalid:** Writing – review & editing, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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